

SSD_Uuid acquisition guide

1. sdk MI_SYS_ReadUuid API read

Read the demo as follows:

```
#include <stdio.h>

#include "mi_sys.h"

#include "mi_common_datatype.h"

int main(void)
{
    MI_U64  u64Uuid ;

    MI_S32 s32Ret = MI_ERR_SYS_FAILED;

    s32Ret = MI_SYS_ReadUuid (&u64Uuid);

    if(!s32Ret)
    {
        printf("uuid: %llx\n", u64Uuid);
    }

    return 0;
```

```
}
```

2. APP reads directly through reg

Read Demo as follows:

```
#include <fcntl.h>

#include <stdio.h>

#include <stdlib.h>

#include <string.h>

#include <sys/mman.h>

#define DBG_INFO printf

#define DBG_ERR printf

#define BIT8 0x100

#define BANK_TO_ADDR32(b) (b<<9)

#define REG_ADDR(riu_base,bank,reg_offset)
((riu_base)+BANK_TO_ADDR32(bank)+(reg_offset*4))

typedef struct
{
    unsigned char *virt_addr;

    unsigned char *mmap_base;

    unsigned int mmap_length;
}MmapHandle;

static unsigned int const page_size_mask = 0xFFF;

MmapHandle* devMemMMap(unsigned int phys_addr, unsigned int length)
```

```

{

    int fd;

    unsigned int phys_offset;

    fd = open("/dev/mem", O_RDWR|O_SYNC);

    if (fd == -1)

    {

        DBG_ERR("open /dev/mem fail\n");

        return NULL;

    }

    MmapHandle *handle = malloc(sizeof(MmapHandle));

    phys_offset =(phys_addr & (page_size_mask));

    phys_addr &= ~(page_size_mask);

    handle->mmap_length = length + phys_offset;

    handle->mmap_base = mmap(NULL, handle->mmap_length ,
    PROT_READ|PROT_WRITE, MAP_SHARED, fd, phys_addr);

    handle->virt_addr = handle->mmap_base + phys_offset;

    DBG_INFO("phys_addr: %#x\n", phys_addr);

    DBG_INFO("virt_addr: %p\n", handle->virt_addr);

    DBG_INFO("phys_offset: %#x\n", phys_offset);

    if (handle->mmap_base == MAP_FAILED)

    {

        DBG_ERR("mmap fail\n");

        close(fd);

        free(handle);

        return NULL;

```

```

    }

    close(fd);

    return handle;
}

int devMemUmap(MmapHandle* handle)
{
    int ret = 0;

    ret = munmap(handle->mmap_base, handle->mmap_length);

    if(ret != 0)
    {
        printf("munmap fail\n");

        return ret;
    }

    free(handle);

    return ret;
}

int main()
{
    unsigned long long uuid;

    /* RIU mapping*/

    MmapHandle *riu_base = devMemMMap(0x1F000000, 0x2B0000);

    /*Configure PAD and Clock here*/

    //chg default dev2,3 pclk from d4 to gpio2

    *(unsigned short*)REG_ADDR(riu_base->virt_addr, 0x20, 0x03) &=
    ~BIT8;

```

```

    uuid = (unsigned long long )(* (unsigned
    short*)REG_ADDR(riu_base->virt_addr, 0x20, 0x16)) |

    ((unsigned long long )(* (unsigned short*)REG_ADDR(riu_base-
    >virt_addr,
    0x20, 0x17)) << 16) |

    ((unsigned long long )(* (unsigned short*)REG_ADDR(riu_base-
    >virt_addr,
    0x20, 0x18)) << 32);

    devMemUmap(riu_base);

    printf("uuid: %llx\n", uuid);

    return 0;

}

```

3. 临时通过debug终端读取

读取方法如下：

1. 读取bank20 offset 03的值，然后把bit8清0

```

/ # /config/riu_r 20 3
BANK:0x20 16bit-offset 0x03
0x0000
/ # /config/riu w 20 3 0
BANK:0x20 16bit-offset 0x03
0x0000
/ # 

```

2. 依次通过读取bank20 offset 16,17,18获取uuid的值

(16: uuid bit0~bit15,17: uuid bit16~bit31,18: uuid bit32~bit47)

如下面读出来uuid的值为0x5b3aff222f5

